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by

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Predicting the Sound Insulation of Walls

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ABSTRACT
Between 1990 and 1998, the author published five conference papers which described the gradual development of a simple theoretical model for predicting the sound insulation of building partitions. The first aim was to extend Sharp's model for cavity walls to cavities without sound absorption. The second aim was to remove the reported over prediction of Sharp's model for cavity walls. The third aim was to explain the five decibel empirical correction in Sharp's model for stud walls. The fourth aim was to produce a more theoretically valid model than Gu and Wang's steel stud wall model. Although the simple theoretical model has been reasonably successful, several concerns have since arisen. This paper describes how these concerns have been addressed and gives the current version of this theoretical model for predicting sound insulation. The theoretical model is compared with a number of experimental measurements and produces reasonable agreement.

1. INTRODUCTION
1.1. Air borne transmission across the wall cavity
Between 1990 and 1998, the author published five conference papers [1,2,3,4,5] which described the gradual development of a simple theoretical model for predicting the sound insulation of building partitions. The starting point was Sharp's research [6,7,8]. The initial work was heavily influenced by Rudder's [9] claim that Sharp's theory predicted less airborne transmission across the cavity than most experimental results. The initial aim was to extend Sharp's theory to cover the case of cavities without added absorption, and to obtain results for cavities with added absorption that agreed with the experimental results presented by Rudder. The sound transmission was calculated by integrating over angles of incidence from zero to the variable limiting angle obtained by Sewell [10] for finite area single leaf walls.

The predictions of this theory were compared with the experimental results in [11]. Reasonable agreement was obtained. The experimental results in [11] were used because they were the average of a number of measurements. However, the sound insulation values are lower than some of the results published in the literature. When the author commenced research in acoustics in Australia, he was told by colleagues that overseas sound insulation measurements were often higher than Australian measurements. The reason was thought to be the use of smaller sample sizes and smaller reverberant rooms overseas. A little detective work has shown that most of the experimental measurements, used by Rudder to reject Sharp's theory for airborne transmission across the cavity, originate from the National Research Council of Canada (NRCC). These NRCC measurements have been respected in Australia because they are at the lower end of overseas results. However research by Warnock [12] has suggested that these NRCC measurements may be effected by various forms of flanking. More recent measurements at NRCC, on a double steel stud wall with a cavity completely filled with sound absorption, have produced results close to Sharp's theory. Most of the earlier NRCC measurements used by Rudder had no studs rather than double studs. This may have had some effect on the results.

The author's results for the sound transmission of a double leaf cavity wall with sound absorbing material in the cavity and no structural connection between the leaves are the same as Sharp's theory if a fixed limiting angle of 61° is used. It is reasonable to suppose that a cavity with a large amount of absorption reduces the range of angles at which sound can propagate effectively across the cavity. Thus in this paper the variable limiting angle for cavity walls is limited to a maximum value of 61°.

Sharp's sound insulation values increase rapidly above the normal incidence mass-air-mass resonance frequency. This effect occurs in the author's theory as a result of the absorption coefficient of the cavity with sound absorbing material increasing as the ratio of the cavity width to wavelength increases. In this paper the maximum cavity absorption coefficient is restricted to be less than or equal to the product of the cavity width with the wave number.

1.2. Rigid stud structure borne transmission across the wall cavity

Sharp successfully modeled the stud borne transmission via wooden studs across the wall cavity using the approach of Cremer et al. [13]. The author followed Fahy's approach [14]. The author found it necessary to include an empirical correction factor of 2 to obtain agreement with Sharp's theory and with experiment.

The author [4] showed that this empirical correction factor of 2 was due to the resonant vibration of the wall leaves. An apparent asymmetry in the wall leaf critical frequencies was explained as being due to total internal reflection when the vibration was transmitted via a stud. However, when the author visited Berlin in 1993, Heckl pointed out that there was still an asymmetry in the damping loss factor of the wall leaves when the critical frequencies are equal.

1.3. Resilient stud structure borne transmission across the wall cavity

The author modeled steel studs as springs. His theory showed that the transmission via steel studs is small compared to the airborne transmission across the cavity, except in the low frequency range near and below the mass-spring-mass resonance frequency.

His theory agreed well with experiment because his airborne transmission results agreed with the experimental results for steel stud walls. This implies that cavity walls with no studs, staggered studs or double studs will perform no better than steel stud walls. The problem is which experimental results should be believed. The lower measured sound insulation results show that the above implication is nearly correct. The higher measured sound insulation values show that the above implication is incorrect.

The author is now inclined to accept the higher measured sound insulation values, and to assume that the lower experimental values were affected by some form of flanking transmission. This implies that his predicted airborne sound transmission values for cavities with absorption must be decreased as discussed in the first subsection. It also implies that his predicted steel stud sound transmission values must be increased. An empirical steel stud structure borne attenuation factor of 0.04 relative to wooden studs has been used in this paper.

2. THIN SINGLE WALLS

The sound transmission coefficient τ of a wall is the ratio of the sound energy transmitted by the wall to the sound energy incident upon the wall. For an infinite, isotropic, uniform thickness plane wall the sound transmission coefficient of a plane wave depends on the angle θ between the direction of propagation of the incident plane wave and the normal to the plane of the wall. To evaluate the diffuse field sound transmission coefficient τ_d it is necessary to average the plane wave sound transmission coefficient $\tau(\theta)$ with appropriate weighting across all angles of incidence,

$$\tau_d = 2\int_0^{\pi/2} \tau(\theta)\cos\theta\sin\theta d\theta. \tag{1}$$

The cos θ term is the cross-sectional area of the plane sound wave that is incident on a unit area of the wall at an angle of incidence of θ to the normal to the wall. The sinθ term is due to the fact that the annulus of solid angle between θ and $\theta + \delta\theta$ is $2\pi\sin\theta\,\delta\,\theta$. The 2 term is a normalization factor which arises from the fact τ_d must be 1 when $\tau(\theta)$ is 1 for all values of θ. Eqn (1) can be rewritten in a number of forms. Use will be made of the following form

$$\tau_d = \int_0^1 \tau(\theta)d(\cos^2\theta). \tag{2}$$

For a thin plane wall with the properties described above Cremer [15] has shown that

$$\tau(\theta) = \frac{1}{\left|1+\dfrac{Z\cos\theta}{2\rho_0 c}\right|^2}, \tag{3}$$

where Z is the bending wave impedance of the wall and $\rho_0 c$ is the impedance of air, being the product of the ambient density ρ_0 and the speed of sound in air c. If the damping and stiffness of the wall are ignored, then Z is equal to $j\,\omega m$ where ω is the angular frequency and m is the mass per unit area of the wall. This means that

$$\tau(\theta) = \frac{1}{1 + a^2 \cos^2\theta}, \tag{4}$$

where

$$a = \frac{\omega m}{2\rho_0 c}. \tag{5}$$

The normal incidence sound transmission coefficient is

$$\tau(0) = \frac{1}{1+a^2} \approx \frac{1}{a^2}, \tag{6}$$

where the approximation applies for the usual case of $a >> 1$.

Evaluating eqn (2) with $\tau(\theta)$ given by eqn (4) gives the diffuse field sound transmission coefficient

$$\tau_d = \frac{\ln(1+a^2)}{a^2} \approx -\tau(0)\ln(\tau(0)), \tag{7}$$

where the approximation is valid for $a >> 1$. Unfortunately τ_d does not agree very well with experimental results. Better agreement with experiment is obtained by limiting the range of angles of incidence over which the sound transmission coefficient is averaged from 0° to a value θ_l between 78° and 85° (see Sharp [7]). The value obtained is called the field incidence sound transmission coefficient and is given by

$$\tau_f = \int_{\cos^2\theta_l}^{1} \frac{d(\cos^2\theta)}{1 + a^2\cos^2\theta} = \frac{1}{a^2}\ln\left(\frac{1+a^2}{1+a^2\cos^2\theta_l}\right), \tag{8}$$

which if $a >> 1$ becomes

$$\tau_f \approx -\tau(0)\ln[\tau(0) + \cos^2\theta_l]. \tag{9}$$

If $\tau(0) << \cos^2\theta_l$ then

$$\tau_f \approx -\tau(0)\ln\left(\cos^2\theta_l\right), \tag{10}$$

and τ_f differs from $\tau(0)$ by a constant factor.

The physical reason for the need to introduce a limiting angle is that the experiments are performed on a finite size wall while the theory assumes a wall of infinite extent. The finite size of the wall means that a bending wave of single wave number becomes a band of wave numbers. For θ near 90° some of the bending wave energy will have wave numbers which are greater than the wave number in air and thus will be unable to radiate. Some of the bending wave energy will have smaller wave numbers for which the predicted sound transmission coefficient is less. Both of these effects mean that to obtain a reasonable answer the upper angle of incidence in the integral has to be limited.

This approach works fairly well for thin single walls but fails badly for cavity walls. The reason is that cavity wall theories are much more sensitive to the value of the limiting angle because they involve the square of the single wall sound transmission coefficient and thus vary with angle of incidence θ as $1/\cos^4\theta$ instead of $1/\cos^2\theta$.

The limiting angle is needed because of the finite size of the wall. Thus it would be expected that the limiting angle should depend on the ratio of a typical wall size dimension to the wavelength of sound, rather than be a constant. It turns out that this is indeed the case. Sewell [10] has shown for a single wall of the type considered in this paper that

$$\tau_f = \tau(0)\ln\left(k\sqrt{A}\right), \tag{11}$$

where k is the wave number of the sound in air and A is the area of the wall. All but the first term in the brackets in Sewell's eqn (54) have been ignored because they are usually insignificant. Comparing eqn (11) with eqn (10) shows that the limiting angle θ_l is given by

$$\cos^2\theta_l = \begin{cases} 0.9 & \text{if } \dfrac{1}{k\sqrt{A}} > 0.9 \\ \dfrac{1}{k\sqrt{A}} & \text{if } 0.9 \geq \dfrac{1}{k\sqrt{A}} \end{cases}. \tag{12}$$

The maximum value of 0.9 is imposed because $\cos^2\theta_l$ has a maximum value of 1. The value of 0.9 is used for compatibility with the cavity wall case where the value of 1 cannot be used.

The referee of this paper has commented that "It is the reviewer's opinion that the limitation of the integration angle that is very important to this paper's approach is possibly outdated by work by for instance Rindel [16]." For single leaf walls the author agrees. However for single leaf walls, the use of the variable limiting angle given by eqn (12) is equivalent to the use of a forced radiation efficiency $\langle\sigma\rangle$ averaged over angles of incidence [17] of

$$\langle\sigma\rangle = \int_0^{\pi/2}\sigma(\theta)\sin\theta d\theta = \frac{1}{2}\ln\left(k\sqrt{A}\right). \tag{13}$$

The variable limiting angle approach is retained in this paper for use with double leaf cavity walls. The author is not aware of any better approach for predicting the sound insulation of double leaf cavity walls.

The bending wave impedance of a thin wall is (see Cremer [15])

$$Z = jm\omega\left[1-\left(\frac{\omega}{\omega_c}\right)^2(1+j\eta)\sin^4\theta\right], \tag{14}$$

where ω_c is the angular critical frequency of the wall (that is, the angular frequency at which the wavelength of free bending waves in the wall equals the wavelength of sound in air), and η is the damping loss factor of the wall. If the damping loss factor and the angular dependence are ignored, it is not surprising to find that Sewell [10] has shown that below the critical frequency the a of eqn (5) becomes

$$a = \frac{\omega m}{2\rho_0 c}\left[1 - \left(\frac{\omega}{\omega_c}\right)^2\right]. \tag{15}$$

To calculate the field incidence sound transmission coefficient below the critical frequency eqns (15), (12) and (8) are used. Above the critical frequency use is made of the equation developed by Cremer [15],

$$\tau_f = \frac{1}{a^2}\frac{\pi}{2\eta}\frac{1}{\dfrac{\omega}{\omega_c} - 1}, \tag{16}$$

where a is given by eqn (5) <u>not</u> eqn (15). Both these methods give indeterminate values at the critical frequency. At the critical frequency use is made of the equation developed by Josse and Lamure [18],

$$\tau_f = \frac{1}{a^2}\frac{\pi}{2\eta}\frac{1}{b}, \tag{17}$$

where a is given by eqn (5) <u>not</u> eqn (15), and b is the ratio of the bandwidth of the filter used in the measurements to the centre frequency of the filter. For a third-octave filter b is equal to 0.2316. Eqn (17) is used for frequencies between half the critical frequency and the critical frequency, if it gives a smaller value than eqn (8). Eqn (17) is also used for frequencies between the critical frequency and twice the critical frequency, if it gives a smaller value than eqn (16).

3. AIR BORNE TRANSMISSION ACROSS THE CAVITY

The normal incidence mass-air-mass resonance angular frequency ω_0 of a cavity wall is given by (see Fahy [14])

$$\omega_0 = \sqrt{\frac{\rho_0 c^2 (m_1 + m_2)}{d m_1 m_2}}, \tag{18}$$

where ρ_0 is the ambient density of air, c is the speed of sound in air, d is the cavity width and m_i is the mass per unit area of the ith leaf of the cavity wall (i = 1 or 2). Below the normal incidence mass-air-mass resonance frequency the cavity wall behaves like a single wall with the same total mass per unit area. Thus its field incidence sound transmission coefficient τ_f at angular frequency ω is given by

$$\tau_f = \frac{1}{a^2}\ln\left(\frac{1 + a^2}{1 + a^2\cos^2\theta_l}\right), \tag{19}$$

where

$$a = \frac{\omega(m_1 + m_2)}{2\rho_0 c}. \tag{20}$$

Above the normal incidence mass-air-mass resonance frequency, the air borne sound transmission across the cavity and the stud borne sound transmission across the cavity are treated separately. The starting point for the prediction of the field incidence sound transmission coefficient for the air borne sound transmission across the cavity is eqn (C-10) of Rudder [9] which is derived using the approach of Mulholland et al. [19]. This equation can be written as follows,

$$\tau(\theta) = \frac{1}{R^2(\theta) + I^2(\theta)}, \tag{21}$$

$$R(\theta) = 1 - A_1 A_2 \left(1 - r^2 \cos 2\beta\right), \tag{22}$$

$$I(\theta) = A_1 + A_2 - r^2 A_1 A_2 \sin 2\beta, \tag{23}$$

$$a_i = \frac{\omega m_i}{2\rho_0 c}\left[1 - \left(\frac{\omega}{\omega_{ci}}\right)^2\right], \tag{24}$$

$$A_i = a_i \cos\theta, \tag{25}$$

$$\beta = kd\cos\theta, \tag{26}$$

where m_i and ω_{ci} are the mass per unit area and angular critical frequency of the ith leaf of the cavity wall (i = 1 or 2), d is the cavity width and r is the reflection factor of the cavity. Some algebra shows that

$$\begin{aligned}\frac{1}{\tau(\theta)} &= \left(\sqrt{(A_1^2+1)\,(A_2^2+1)} - r^2 A_1 A_2\right)^2 + \\ &4r^2 A_1 A_2 \sqrt{(A_1^2+1)\,(A_2^2+1)}\,\sin^2\left(\beta - \frac{1}{2}\arctan\left(\frac{A_1 + A_2}{A_1 A_2 - 1}\right)\right).\end{aligned} \tag{27}$$

$\tau(\theta)$ has its maximum values at resonance when the sin argument in the second term is equal to an integer multiple of π. The mass-air-mass resonance occurs when the sin argument is equal to zero. Above the normal incidence mass-air-mass resonance frequency, there is always an angle of incidence at which the mass-air-mass resonance occurs. Except for values of the cavity reflection factor r which are very close to one, the "bandwidth" of the resonance in terms of angle of incidence is broad. This means that $1/\tau(\theta)$ in the integral over angle of incidence can be approximated by ignoring the second term in eqn (27). Using the fact that A_i is usually large compared with one, the first term of eqn (27) can be approximated to obtain

$$\tau(\theta) = \frac{1}{(q+px)^2}, \tag{28}$$

where

$$q = \frac{1}{2}\left(\frac{a_2}{a_1} + \frac{a_1}{a_2}\right), \tag{29}$$

$$p = a_1 a_2 \alpha, \tag{30}$$

$$\alpha = 1 - r^2, \tag{31}$$

and

$$x = \cos^2\theta. \tag{32}$$

α is the absorption coefficient of the cavity.

The field incidence sound transmission coefficient is

$$\tau_f = \int_{\cos^2\theta_l}^{1} \frac{dx}{(q+px)^2} = \frac{1-\cos^2\theta_l}{(q+p\cos^2\theta_l)\,(q+p)}. \tag{33}$$

To make eqn (33) agree better with experimental results and Sharp's theory [7] for cavity walls with sound absorbing material in the cavity, it has been found necessary, as discussed in the introduction, to limit θ_l to values less than or equal to 61°. Because θ_l has to be greater than zero in order for eqn (33) to produce positive results, $\cos^2\theta_l$ is limited to be less or equal to 0.9. Thus for cavity walls eqn (12) is replaced with

$$\cos^2\theta_l = \begin{cases} 0.9 & \text{if } \dfrac{1}{k\sqrt{A}} > 0.9 \\ \dfrac{1}{k\sqrt{A}} & \text{if } 0.9 \geq \dfrac{1}{k\sqrt{A}} \geq \cos^2 61°. \\ \cos^2 61° & \text{if } \cos^2 61° > \dfrac{1}{k\sqrt{A}} \end{cases} \tag{34}$$

Because the absorption of any absorbing material in the cavity is limited by the depth of the cavity, the maximum value of the absorption coefficient α is also limited.

$$\alpha \leq \begin{cases} kd & \text{if } kd < 1 \\ 1 & \text{if kd} \geq 1 \end{cases}. \tag{35}$$

This limitation also makes the theory agree with Sharp's theory [7] for cavity walls with sound absorbing material in the cavity. For cavities without absorbing material, the use of an absorption coefficient between 0.10 and 0.15 is suggested.

Eqn (33) is only valid between the normal incidence mass-air-mass resonance frequency and the lower of the critical frequencies of the two cavity leaves. For frequencies between 2/3 of the mass-air-mass resonant frequency and the mass-air-mass resonant frequency the sound reduction index is interpolated in the logarithmic frequency domain using the sound reduction index calculated from eqn (19) at 2/3 of the mass-air-mass resonant frequency and the sound reduction index calculated from eqn (33) at the mass-air-mass resonant frequency.

When the reflection factor r is very close to one a method similar to that used by Cremer [15] in deriving eqn (16) can be used. The idea is that the main contribution to the integral over $\cos^2\theta$ comes from angles of incidence close to the oblique mass-air-mass resonance which occurs when the argument of the sin term in eqn (27) is zero. Under these circumstances the sin function can be approximated by its argument and the other terms replaced by their value at the oblique mass-air-mass resonance angle. The lower limit of integration of 0 in eqn (2) which has been restricted to $\cos^2\theta_l$ because of the discussion just before eqn (8) is now extended to $-\infty$. The upper limit of integration in eqn (2) is extended from 1 to $+\infty$. Use is made of the fact that A_1 and A_2 are usually very much larger than one. This approach yields the following formula,

$$\tau_f = \frac{\pi\sqrt{s}}{a_1 a_2 r\left(\sqrt{t}+\frac{1}{\sqrt{t}}\right)\left[\sqrt{(st+1)\left(\frac{s}{t}+1\right)}-r^2 s\right]}, \tag{36}$$

where

$$s = \frac{M_1+M_2}{4\rho_0 d}, \tag{37}$$

$$t = \frac{M_1}{M_2}, \tag{38}$$

and

$$M_i = m_i\left[1-\left(\frac{\omega}{\omega_{ci}}\right)^2\right]. \tag{39}$$

This formula is not of much practical use because r will always be somewhat less than 1 because of sound transmission through the leaves, air absorption and boundary absorption of the leaves. However it does agree fairly well with the numerical calculations of London [20] for a limp, lossless cavity wall above the normal incidence mass-air-mass resonance frequency. Like eqn (33), eqn (36) is only valid between the normal incidence mass-air-mass resonance frequency and the lower of the critical frequencies of the two cavity leaves.

When the frequency is equal to or greater than the lower of the two critical frequencies, a method similar to that used by Cremer [15] in deriving eqn (16) can be used. The idea is that the main contribution to the integral over $\cos^2\theta$ comes from angles of incidence close to the coincidence angles. Under these circumstances the $\sin^4\theta$ term in eqn (14) for each leaf of the cavity wall can be approximated by a linear expansion about $\sin\theta_{ci}$ where θ_{ci} is the coincidence angle for each leaf of the cavity wall, and the other terms can be approximated by their values at the coincidence frequencies. The lower limit of integration of 0 in eqn (2) which has been restricted to $\cos^2\theta_l$ because of the discussion just before eqn (8) is now extended to $-\infty$. The upper limit of integration in eqn (2) is extended from 1 to $+\infty$. This approach yields the following formulae,

$$B_i = \frac{\omega m_i}{2\rho_0 c}, \tag{40}$$

$$\xi_i = \sqrt{\frac{\omega}{\omega_{ci}}}, \tag{41}$$

$$n = \eta_1\xi_2 + \eta_2\xi_1, \tag{42}$$

$$v = 4(\xi_1 - \xi_2), \tag{43}$$

$$\tau_f = \frac{\pi(\xi_1 + \xi_2)n}{4B_1^2B_2^2\eta_1\eta_2\xi_1\xi_2(n^2 + v^2)\alpha^2}, \tag{44}$$

where η_i are the damping loss factors of the ith leaf. Because of the assumptions made, eqn (44) is only valid if the critical frequencies of the two leaves are not too different. Eqn (33) is ill defined at the critical frequencies of the panels. Thus eqn (44) is used for all frequencies which are greater than 0.9 times the lower of the two critical frequencies.

4. STUD BORNE TRANSMISSION ACROSS THE CAVITY

The stud borne sound transmission theory was originally developed by Sharp [6,7], Sharp *et al*. [8] and Cremer *et al*. [13]. Fahy's [14] approach to the theory was followed, but the cavity leaves were allowed to have different masses per unit area and different critical frequencies. It was also assumed that the studs had a mechanical compliance of C_M where $C_M = 0$ gave the rigid stud case of previous authors. Gu and Wang [21] have also treated the case of resilient studs but their formulae are different to those in this paper and are not obviously an extension of Sharp's formulae as is the case with those in this paper.

This approach produces a value of $\tau(\theta)$ that varies with angle of incidence θ as $1/\cos\theta$. Thus from eqn (1) we need to evaluate an integral of the form

$$2\int_0^{\pi/2} \sin\theta d\theta = 2. \tag{45}$$

Strictly speaking the upper limit of the integral should be θ_l, but the value of eqn (45) will not be changed significantly by setting the upper limit to $\pi/2$. The field incidence sound transmission coefficient is

$$\tau_f = \frac{64\rho_0^2 c^3 D}{[g^2 + (4\omega^{3/2} m_1 m_2 c C_M - g)^2] b\omega^2}, \tag{46}$$

where

$$g = m_1 \omega_{c2}^{1/2} + m_2 \omega_{c1}^{1/2}. \tag{47}$$

b is the spacing between the studs and D is a factor to account for the effects of resonant vibration.

Eqn (46), without the factor D, is only a lower limit because it does not include resonant radiation since it assumes an infinite wall and that the studs do not interact. It also assumes that the bending waves induced in the wall by the incident sound are incident normally to the studs. To improve the agreement between theory and experiment and between eqn (46) and Sharp's formula, D was set to the empirically constant value of 2. This difference arises because Sharp assumed that the correction factor for averaging over angle of incidence was the square of his factor for a single leaf wall, namely 1.9^2, while our theory shows that according to eqn (45) the factor should be only 2. Sharp's formula also includes an empirical 5 dB correction factor which is compensated for in our theory by the square of the ratio of the sum of the wall impedances to the value of one of them, which in the case of identical leaves gives a factor of 4. Surprisingly one of Sharp's [7] formulae includes both the 5 dB correction factor and the impedance ratio, but it appears that he has not compared this formula with experiment. Formula (46) with D equals 2, is like formulae (33) and (36), only valid between the normal incidence mass-air-mass resonance and the lower of the critical frequencies of the two leaves.

Treating steel studs as resilient line connections does not agree with experimental results. For steel studs, it is recommended that the stud compliance C_M be set to zero in eqn (46) and that eqn (46) be multiplied by a stud attenuation factor in the range from 0.02 to 0.2. A stud attenuation factor of 0.04 is used to compare with experimental results in this paper.

The total field incidence sound transmission coefficient of a cavity wall is determined by adding eqn (33) or eqn (44) to eqn (46).

5. EFFECTS OF RESONANT VIBRATION

In the previous section, formulae for the sound transmission of cavity walls via the studs for frequencies below the minimum of the critical frequencies of the wall panels were presented. These formulae contain an empirical constant of two. This constant was included to make the formulae agree better with experimental results and Sharp's [7] earlier work. In this low frequency range, the theory on which the formulae were based only included the nonresonant response and radiation of the panels. This section replaces this empirical constant of two with a theoretical expression which includes the effects of the resonant response and radiation of the wall panels and extends the theory to frequencies above the minimum of the critical frequencies of the wall panels.

Measurements by Crocker and Price [22] showed that the response of a single aluminium panel is more than 10 dB greater than the forced nonresonant mass law response of the panel. Because of this, it was decided to calculate the effects of the resonant response of a gypsum plaster board panel. Eqn 48 of [22] can be rearranged to show that the ratio of the energy of the *i*th panel due to its resonant response to the energy due to its nonresonant forced mass law response is given by

$$e_i = \frac{\pi\omega_{ci}\sigma_i}{4\omega\eta_i}, \tag{48}$$

where ω is the angular frequency, ω_{ci} is the critical angular frequency of the *i*th panel, σ_i is the single sided radiation efficiency of the *i*th panel and η_i is the total loss factor of the *i*th panel which is equal to the sum of the internal loss factor of the *i*th panel $\eta_{int,i}$ and twice (to take account of both sides of the panel) its single sided radiation loss factor $\eta_{rad,i}$. Thus $\eta_i = \eta_{int,i} + 2\eta_{rad,i}$. The single sided radiation loss factor is related to the single sided radiation efficiency by $\eta_{rad,i} = \sigma_i\rho_0 c/(m_i\omega)$, where ρ_0 is the ambient density of air, c is the speed of sound in air and m_i is the mass per unit area of the *i*th panel. The ratio of the sum of the energies of the resonant and nonresonant responses to the energy of the nonresonant response of the *i*th panel for airborne excitation is $1 + e_i$.

Eqn 2.98 of Fahy [14] shows that the ratio of the sound power radiated by the resonant vibration to the sound power radiated by the forced near field vibration for a line source on the *i*th panel is given by

$$r_i = \frac{\sigma_i}{2\eta_i}\sqrt{\frac{\omega_{ci}}{\omega}}. \tag{49}$$

The ratio of the sum of the sound powers radiated by the resonant and nearfield vibrations to the sound power radiated by the nearfield vibration for a line source on the *i*th panel is $1 + r_i$. The ratio of the combined effect of the resonant and nonresonant response and radiation to the effect of the nonresonant response and radiation is given by the product $D = (1 + e_1)\,(1 + r_2)$. This product replaces the empirical factor of two that was used in the previous section, because the theory on which the previous section is based only models the nonresonant excitation and radiation.

The corrected versions of Maidanik's formulae for the single sided radiation efficiency given by Vér and Holmer [23] have been used in this paper. However the maximum value of the radiation efficiency has been limited to the value one. Previous work has shown that this assumption works well for the predicting the sound transmission of third octave bands of noise.

5.1. Comparison with empirical factor of two

Figure 1 compares $D = (1 + e)(1 + r)$ with the empirical factor of two for a 16 mm thick gypsum plaster board panel measuring 3.05×2.44 m with an internal loss factor of 0.04.

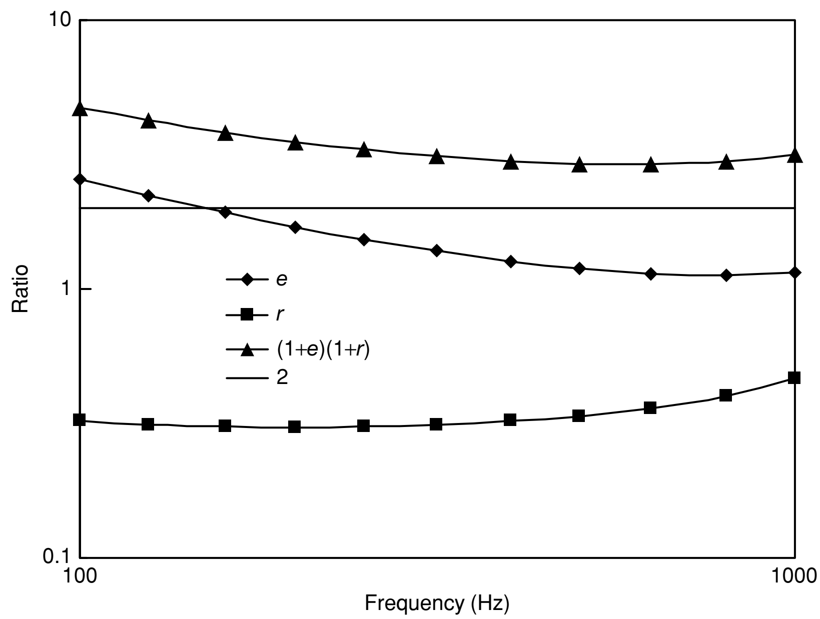

Figure 1. Ratio of resonant to nonresonant response and radiation for 16 mm gypsum plaster board.

It also shows the values of e and r. It can be seen that, in the mid-frequency range, e is of the order of 1 and r is of the order of 0.3. Thus $(1 + e)(1 + r)$ is of the order of 3 in the mid frequency range. This is higher than the empirical correction factor of two, but is of the same order of magnitude. It was thought that the empirical factor of two was due to the resonant radiation of the panel on the receiving room side. Figure 1 shows that the effect of the resonant radiation is generally significantly smaller than two for the gypsum plaster board walls under consideration. This figure also shows that the empirical correction factor of two is mainly due to the resonant response of the panel on the sending room side.

5.2. Apparent asymmetry

The use of the factor $D = (1 + e_1)(1 + r_2)$ introduces an apparent asymmetry into the formula for sound transmission. Such asymmetry would mean that for a stud wall with panels with different critical frequencies, the calculated sound transmission would depend on the direction of transmission through the wall. This arises because the theory used for the transmission of vibration via the line connections of the studs assumes that the bending waves in the panel are incident normally to the line of the stud. For identical panels, calculations have shown that the normal incident value is a good approximation to the value averaged with appropriate weighting over all angles of incidence. For panels with different critical frequencies the situation is much more complicated. "Total internal reflection" will occur for energy flowing from the panel with the higher critical frequency to the panel with the lower critical frequency for angles of incidence greater than a certain angle. To avoid "total internal reflection" we must perform our calculations for the vibrational energy flowing from the panel with the lower critical frequency to the panel with the higher critical frequency. To ensure this we specify that we must number the panels such that $\omega_{c1} << \omega_{c2}$, and use the correction $D = (1 + e_1)(1 + r_2)$. This requirement removes the apparent asymmetry in critical frequency. If this numbering requirement of the panels is not satisfied, a correction factor for the effect of "total internal reflection" needs to be included. Applying this extra correction factor produces the same final result as numbering the panels correctly.

As pointed out in the introduction, there was still an asymmetry in the damping loss factor of the wall leaves when the critical frequencies are equal. In most cases, when the critical frequencies are equal, the wall leaves will be the same and have the same damping loss factor. If this is not the case, the use of the average damping loss factor for both leaves is recommended.

5.3. Theoretical formula

In the light of the results reported in this section

$$D = \left(1 + e_1\right)\left(1 + r_2\right), \tag{50}$$

where the panels have been numbered so that $\omega_{c1} \leq \omega_{c2}$ applies.

6. COMPARISON WITH EXPERIMENT

In this section the theoretical equations are compared with experimental results for a number of gypsum plaster board walls. The gypsum plaster board was assumed to have a density of 770 kg/m^3, a product of surface density with the critical frequency of 31000 kg/m^2/s and an internal damping loss factor of 0.04.

Figure 2 shows a comparison of theory with experiment for a single leaf wall of 13 mm gypsum plaster board. Three experimental results are shown. The first is for no studs, while the second and third are for 50×100 mm wooden studs spaced at 400 or 600 mm centers respectively. The experimental results show that the studs do not make any significant difference, while the theory slightly but significantly overestimates the experimental results in the lower frequencies. The experimental results were measured by the National Research Council of Canada, and obtained from DuPree's [24] catalogue. The no stud results are the average of three separate measurements.

The theory is compared with experimental results for a 40 mm double steel stud 16 mm gypsum plaster board cavity wall with cavity absorption in figure 3. There was a 10 mm gap between the separate studs. This is a case where there is no vibration connection between the two leaves of the wall (except possibly at the edges) and hence only the air borne cavity wall transmission equations are involved. The value one was used for the cavity absorption for cavity walls with cavity absorption in this paper since the experimental results show little dependence of sound insulation on the type or the thickness of the cavity absorption. The experimental measurements in figure 3 were measured by the National Research Council of Canada (NRCC). The last of the walls measured had no studs.

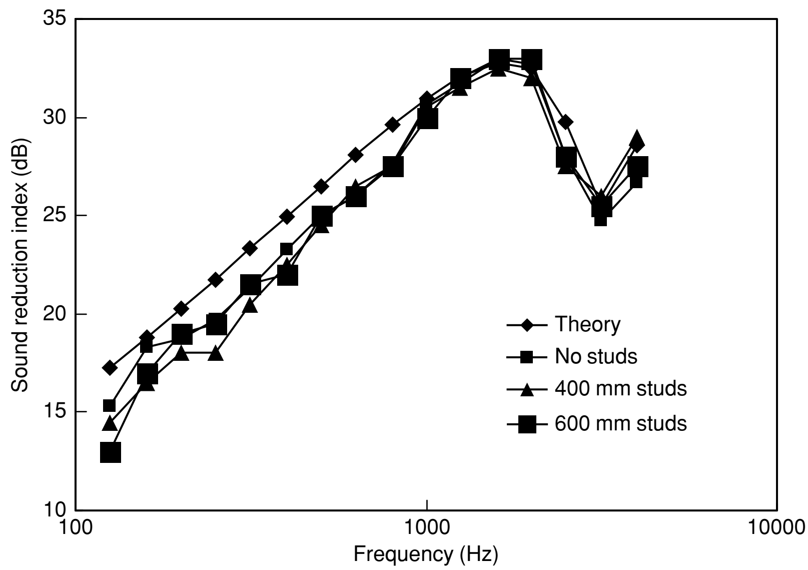

Figure 2. 13 mm gypsum plaster board.

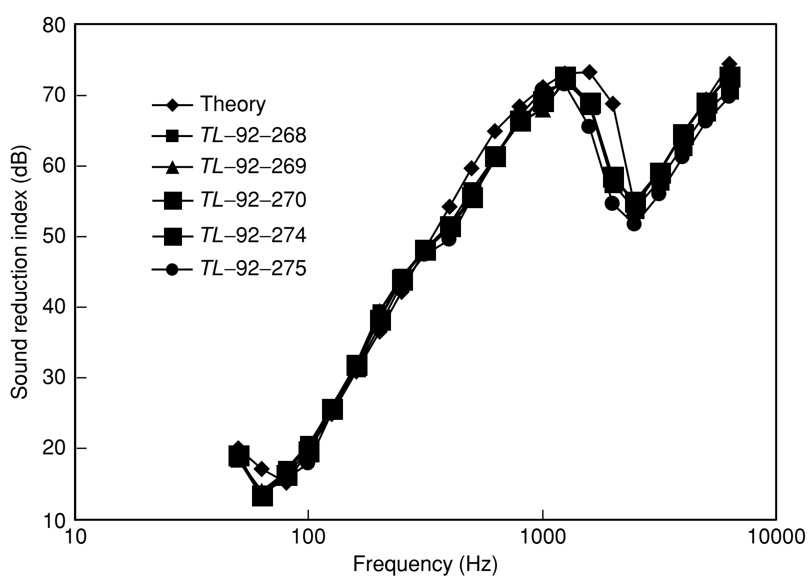

Figure 3. Double stud 16 mm gypsum plaster board cavity wall with cavity absorption.

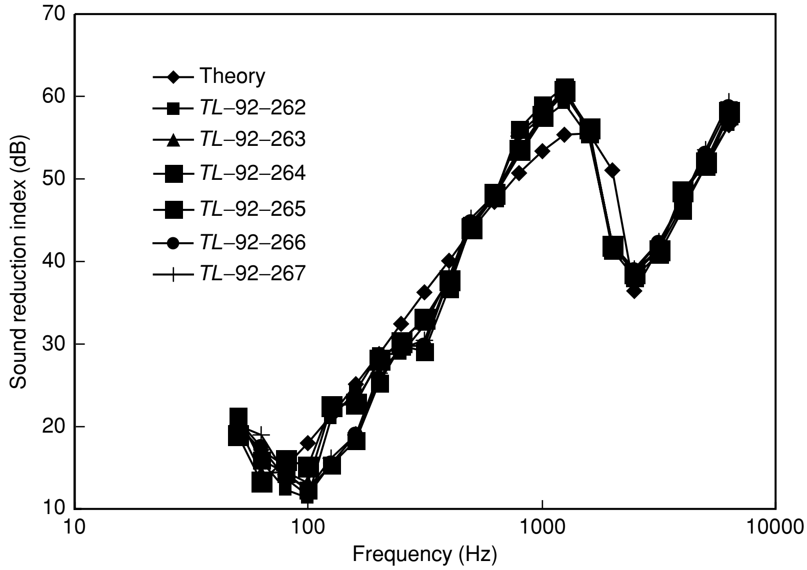

Figure 4. 16 mm gypsum plaster board cavity wall without cavity absorption.

The last three experimental results in figure 4 are for the same construction as the results in figure 3, except that there is no sound absorbing material in the cavity. In the first three experimental results, the double 40 mm steel studs with a 10 mm gap are replaced with 90 mm steel studs. These 90 mm steel stud results have been included, because surprisingly they are as good as or better than the double 40 mm steel stud results. The theoretical curve does not include stud borne transmission. Another result for the case with no studs, which is not included here, produced lower results. Presumably this is because the studs help inhibit the oblique propagation of sound in the cavity in this case without sound absorption in the cavity.

Figure 5 shows the case for resilient studs. It is for a steel stud 13 mm gypsum plaster board cavity wall with cavity absorption. The stud width is 90 mm and the stud spacing is 610 mm. The stud attenuation factor used to obtain the theoretical results is 0.04. The theoretical values in the region of the peak just below the critical frequency depend on the value of this stud attenuation factor. These experimental results are from the NRCC [25].

The results for a wooden stud 13 mm gypsum plaster board cavity wall with cavity absorption are shown in figure 6. The studs are 90 mm spaced at 406 mm centers. The experimental results are also from the NRCC [25]. The theoretical sound transmission is determined mainly by the transmission across the non-resilient studs. The disagreement between theory and experiment is surprising because the theoretical results are close to Sharp's [6,7,8] theory. Sharp's theory has been widely considered to provide reasonable agreement with the experimental results for wooden stud gypsum plaster board cavity walls with cavity absorption. The NRCC [25] experimental results are certainly different from previous experimental results [11].

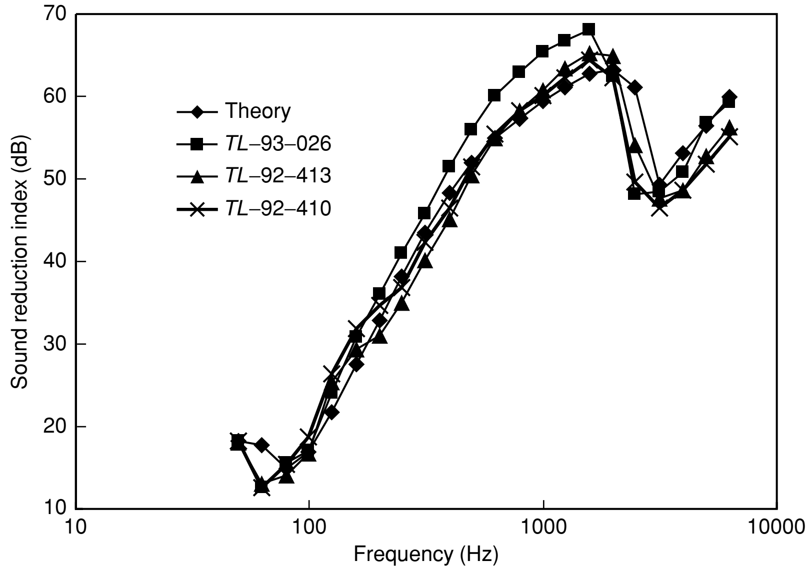

Figure 5. Steel stud 13 mm gypsum plaster board cavity wall with cavity absorption.

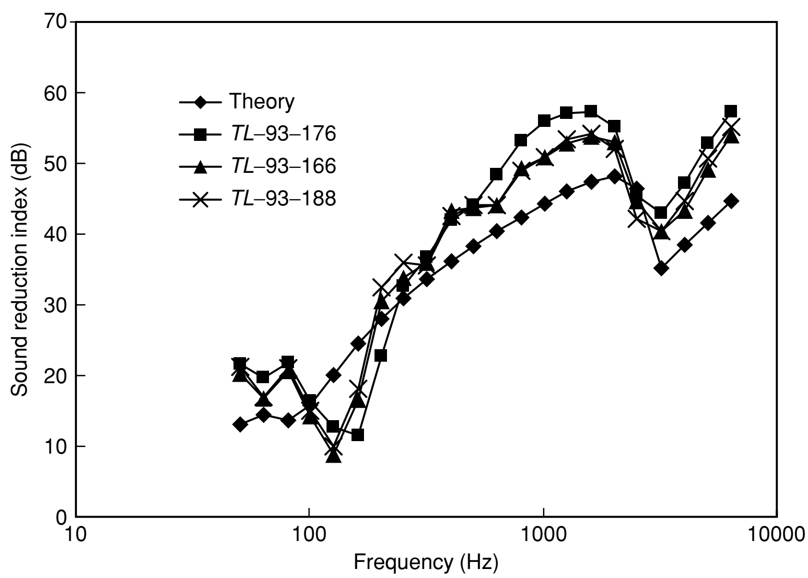

Figure 6. Wooden stud 13 mm gypsum plaster board cavity wall with cavity absorption.

7. CONCLUSION

The current version of the prediction method presented in this paper gives fairly reasonable agreement with the experimental results for the sound transmission loss of gypsum plaster board walls. Hongisto [26] reported that the previous version of the prediction method described in this paper was recommended for rigid and flexible stud walls with sound absorbing material in the wall cavity, but that it under-estimated the sound insulation of cavity walls without studs, both with and without sound absorbing material in the wall cavity. Figures 3 and 4 show that the changes to the prediction method introduced in this paper also make it useful in these last two situations.

The previous version of the prediction method presented in this paper was made more widely available than its conference paper publication allowed, by its inclusion in both the second [27] and third editions of Bies and Hansen's textbook. The author believes that the changes introduced to the prediction method in this paper will make it even more widely useful. Its publication in this journal will certainly make it more widely and easily available.

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